

Day 3 — Two-Digit $\times$ Two-Digit Near 100 (Above)

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute 103×104 using excesses.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ādhikya* (IAST: ādhikya; lit. “excess (number above base)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Above-100 variant.** Use excesses instead of deficits.
 - $103 = 100 + 3 \rightarrow$ excess +3.
 - $104 = 100 + 4 \rightarrow$ excess +4.
2. **The two-step rule for above-base.**
 - **Left part:** cross-add. $103 + 4 = 107$ (or $104 + 3 = 107$).
 - **Right part:** product of excesses. $3 \times 4 = 12$.
 - Combine: **10712**.
3. **Worked examples.**
 - $103 \times 104 = 107 \mid 12 = \mathbf{10712}$.
 - $110 \times 105 = 115 \mid 50 = \mathbf{11550}$.
 - $102 \times 109 = 111 \mid 18 = \mathbf{11118}$.
4. **Algebra (above-base).** $(B+a)(B+b) = B(B+a+b) + ab$. Cross-ADD = $B+a+b$. Product of excesses = ab .
5. **Symmetry recognition.** Below-base: cross-SUBTRACT. Above-base: cross-ADD. Both: product-of-difference-from-base. “*This symmetry hints at the general pattern — tomorrow we’ll see the mixed case.*”

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Two-Digit $\times$ Two-Digit Near 100 (Above).

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).